AWS VPC Peering with VPC-A to VPC-B Host

1. Project Overview

This project demonstrates communication between two Amazon Virtual Private Clouds (VPCs) using VPC Peering.

It uses a VPCs host in a public subnet to provide SSH access to private EC2 instances. The public - private instances communicate using private IP addresses

2. Objectives

* Create two VPCs with non-overlapping CIDR blocks.
* Configure public and private subnets.
* Connect two VPCs using VPC Peering.
* Configure route tables for inter-VPC communication.
* Use a public subnet host to access private EC2 instances.
* Configure security groups to control SSH access.
* Test connectivity using private IP addresses.

3. Architecture

VPC-A

* CIDR: `10.0.0.0/16`
* Public subnet: `10.0.0.0/24`
* Private subnet: `192.168.0.0/24`
* Internet Gateway: Attached
* public-subnet-vpc-1-instance: Public subnet
* Private-EC2-VPC-A: Private subnet

Private-VPC-2

* CIDR: `192.168.0.0/16`
* private-subnet-vpc-2(B): `10.0.1.0/24`
* private-ec2-vpc-2-instance: Private subnet

Connectivity

VPC1 and VPC2 communicate through a VPC Peering Connection.

The public host provides SSH access to the private instances. The private instances use private IP addresses for communication.

4. AWS Services Used

* Amazon VPC
* Amazon EC2
* VPC Peering
* Internet Gateway
* Route Tables
* Public and Private Subnets
* Security Groups
* SSH

5. Network Configuration

Resource	CIDR / Configuration
VPC1	10.0.0.0/16
VPC1 Public Subnet	10.0.0.0/24
VPC1 Private Subnet	192.168.0.0/24
VPC2	192.168.0.0/16
VPC2 Private Subnet (current)	10.0.1.0/24

Each `/24` subnet contains 256 total IPv4 addresses, of which AWS reserves 5.

6. Routing

After correcting the subnet CIDRs and establishing an active peering connection, configure these routes.

VPC1 Public Route Table
10.0.0.0/16 → local
0.0.0.0/0 → Internet Gateway
192.168.0.0/16 → VPC Peering Connection

VPC1 Private Route Table
10.0.0.0/16 → local
192.168.0.0/16 → VPC Peering Connection

VPC2 Private Route Table
192.168.0.0/16 → local
10.0.0.0/16 → VPC Peering Connection

7. Security Configuration

SSH access to the public bastion instance should be restricted to the administrator's public IP address.
Private EC2 instances should not have public IP addresses.
SSH access to private instances should be restricted using appropriate security group rules.
Private network communication should use private IP addresses.
Private EC2 instances should not receive direct inbound internet access.

8. Testing

Laptop-to-Bastion Connectivity: Successfully connected from the local laptop to the public EC2 bastion instance.

Amazon EC2 Connect: Successfully accessed the EC2 instances using Amazon EC2 Connect.

VPC1 Connectivity: Successfully tested connectivity to the private EC2 instance within VPC1 using ping requests.

VPC2 Connectivity: Successfully tested connectivity to the private EC2 instance within VPC2 using ping requests.

Cross-VPC Connectivity: Successfully verified communication between instances in both VPCs through the VPC Peering connection using their private IP addresses.

All tests were successful, confirming that the configured routing, security group rules, and VPC Peering connection support the intended private network communication.

9. Learning Outcomes

Understanding VPCs, subnets, and CIDR blocks.
Configuring VPC Peering and route tables.
Understanding public and private EC2 instances.
Configuring SSH access through a bastion host.
Troubleshooting AWS network connectivity.

10. Conclusion

This project demonstrates how AWS VPC Peering enables private communication between two VPCs. A bastion host provides a controlled administrative entry

point, while private EC2 instances communicate through private IP addresses without requiring public IP addresses.